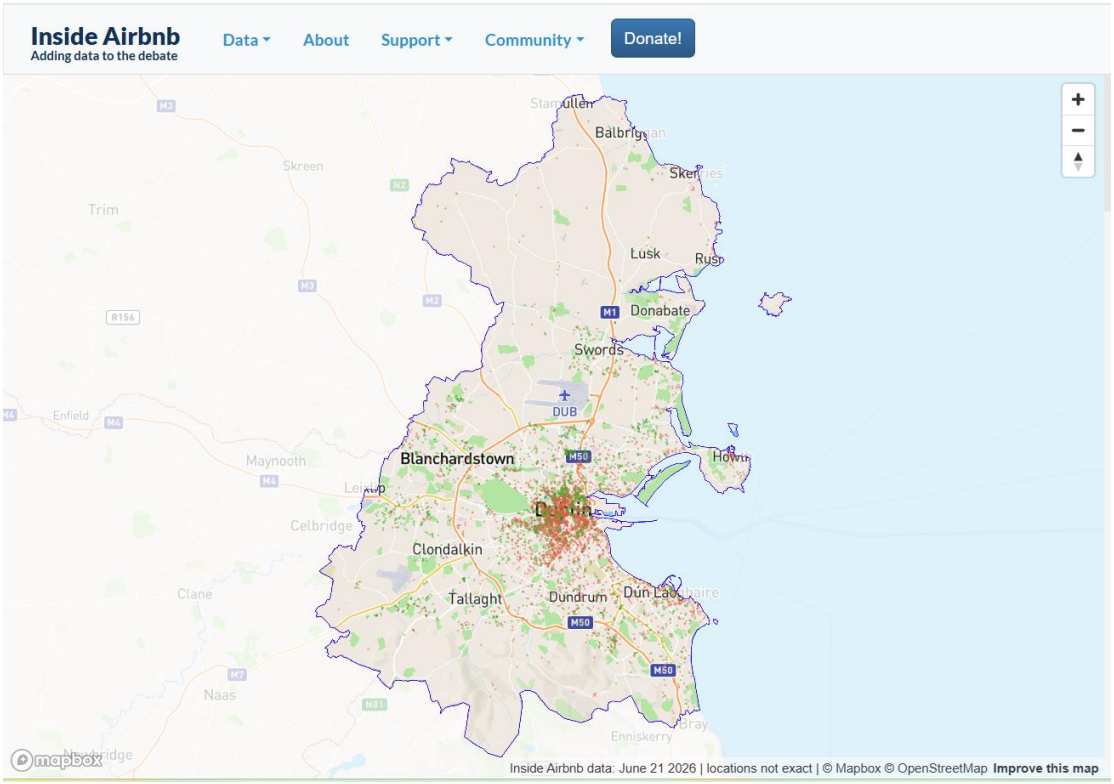


Figure: InsideAirbnb's public Dublin listings map (data snapshot dated 21 June 2026). Source: insideairbnb.com/Dublin .

1. Business question

A host-support team at Airbnb wants to understand what are the significant predictors of successful bookings at a listing and whether the predictors identified by past research conducted across different cities, also hold in Dublin.

2. The broader business problem

Airbnb does not release booking data, so researchers often use proxy measures of successful bookings such as the count of guest reviews a listing receives (Zhang et al., 2018, Biswas et al., 2020). Biswas et al. (2020) show, across ten cities, that host-generated features such as superhost status, host tenure, amenities and guest-generated features such as review scores, and sentiment can determine this count. Sengupta et al. (2021) extend the question to 22 cities, including Dublin, and report that a large share of listings receive no bookings at all, with superhost status, host response time and guest communication as the strongest predictors everywhere.

This option puts the same question to Dublin's own listings: does a single city's live data support what past studies found across ten and twenty-two cities?

3. Data

For Dublin and another city, download the following pair of files from InsideAirbnb website: https://insideairbnb.com/get-the-data/

- **listings.csv**, one row per listing with host, pricing and amenity attributes; and
- **reviews.csv**, one row per individual guest review, with fields listing_id, id, date, reviewer_id, reviewer_name and comments.

Field definitions and data types for both files should be taken from the official Inside Airbnb Data Dictionary:
https://docs.google.com/spreadsheets/d/1iWCNJcSutYqpULSQHlNyGInUvHg2BoUGoNRIGa6Szc4/edit?usp=sharing
which students should cite directly rather than inferring column meanings from the raw file. The schema has changed across years, so the version being used should be reported.
Because reviews.csv is recorded at the review level, it must be aggregated to one row per listing "count of reviews", average comment length, and so on ….. before it can be joined to listings.csv using listings.csv.id = reviews.csv.listing_id.
Each group should report the download date, the row count of each raw file, and the row count remaining after aggregation and joining, since these figures change with every quarterly refresh and cannot be produced from archives.
The resulting listing-level analysis dataset should fall between approximately 5,000 and 50,000 rows - large enough to support stable count-regression estimates, but small enough to clean, join and model on a standard laptop.
Dublin's own listings.csv should sit comfortably within this range; a group that adds a second, larger city for the extension described below should subsample its listings to stay within the same bounds.
A group aiming for a stronger submission may repeat the exercise for a second InsideAirbnb city drawn from Sengupta et al.'s 22-city sample, for example Amsterdam, Barcelona or Copenhagen and compare its results against the Dublin findings and against the published cross-city table.

4. Variable definitions

Field (source file)	Definition and role
host_is_superhost (listings.csv)	Superhost status
host_response_time (listings.csv)	Host responsiveness
host_total_listings_count (listings.csv)	Total listings owned by the host
accommodates (listings.csv)	Accommodation capacity
amenities (listings.csv)	JSON field; parsed and counted to give number of amenities
price (listings.csv)	Nightly price
review_scores_rating, review_scores_communication, review_scores_cleanliness (listings.csv)	Guest review scores
instant_bookable (listings.csv)	Booking policy
comments, aggregated by listing_id (reviews.csv)	Engineered per listing: comment count (cross-checked against number_of_reviews), average

Field (source file)	Definition and role
	comment length, and share of comments with positive sentiment
first_review, last_review (listings.csv)	Used to construct the spread, in days, between a listing's first and most recent review
number_of_reviews (listings.csv)	Dependent variable proxy for successful bookings

5. Modelling approach

Primary model. Consider number_of_reviews as a count outcome, using Poisson or Negative Binomial regression. Students should report the proportion of listings with zero reviews and the variance-to-mean ratio and use this evidence as both papers do, to justify their choice between Poisson and Negative Binomial.

Acceptable alternative. Students may instead fit a linear regression model with a modified dependent variable such as log(1 + number_of_reviews), provided the log transformation is justified on the grounds of the skewness property of the count variable, and the choice is stated explicitly as a simplification relative to the standard Poisson/Negative Binomial approach used in the relevant literature.

6. Note on rigour

Aggregating review-level data to listing level before the join operation is a complex data-cleaning exercise, applied here to Airbnb guest comments. Parsing the amenities field out of its JSON representation is a further cleaning task of the same kind. Justifying Negative Binomial over Poisson, or defending the log-linear alternative, requires students to examine the data dispersion rather than cite a general rule. Because both source files refresh quarterly, no group's dataset, and therefore no group's results, will match those of a previous year.

References:

Biswas, B., Sengupta, P., & Chatterjee, D. (2020). Examining the determinants of the count of customer reviews in peer-to-peer home-sharing platforms using clustering and count regression techniques. *Decision Support Systems*, *135*, 113324.

Sengupta, P., Biswas, B., Kumar, A., Shankar, R., & Gupta, S. (2021). Examining the predictors of successful Airbnb bookings with Hurdle models: Evidence from Europe, Australia, USA and Asia-Pacific cities. *Journal of Business Research*, *137*, 538-554.

Zhang, L., Yan, Q., & Zhang, L. (2018). A computational framework for understanding antecedents of guests' perceived trust towards hosts on Airbnb. *Decision Support Systems*, *115*, 105-116.

Students will also find numerous articles on related topics that have been published in tourism and hospitality journals such as *Tourism Management*, *Annals of Tourism Research* and *International Journal of Hospitality Management*.